

# **AI-Based Real-Time Stress and Trauma Assessment Module for NHAA (14566)**

Problem Statement and Proposed Solution - Revision 5 (Final)

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# **1. Problem Statement**

## **1.1 Background**

Victims and complainants belonging to Scheduled Castes and Scheduled Tribes who approach the National Helpline Against Atrocities (14566), the Integrated Portal, chatbot, mobile application, IVRS, or other digital platforms often experience severe emotional distress arising from caste-based discrimination, violence, rape, gang rape, murder of family members, social boycott, displacement, threats, and prolonged legal proceedings. Presently, there is no standardized mechanism for assessing the psychological condition and vulnerability of victims at the time of first contact with authorities.

## **1.2 The Problem**

Design and develop an AI-enabled Real-Time Stress and Trauma Assessment Module that can assess the psychological stress, trauma, fear, anxiety, and vulnerability levels of victims/complainants interacting through NHAA (14566), the Integrated Portal, chatbot, IVRS, mobile application, or any other approved digital interface.

## **1.3 Expected Solution Requirements**

- Analyse voice interactions, speech patterns, pauses, pitch variation, emotional indicators, and textual narratives
- Use Natural Language Processing (NLP), Speech Analytics, and Emotion AI to identify signs of trauma and distress
- Generate a Stress Vulnerability Index (SVI) on a predefined scale
- Categorize victims into Low, Moderate, High, and Critical Risk categories
- Detect indicators of severe trauma, fear, depression, suicidal ideation, intimidation, social isolation, and extreme vulnerability
- Automatically recommend counselling, legal aid, medical assistance, police intervention, witness protection, or emergency support based on risk level
- Support multilingual interactions, including major Indian languages and dialects
- Maintain privacy, informed consent, confidentiality, and ethical AI standards

## **1.4 Stakeholders**

Department of Social Justice and Empowerment; National Helpline Against Atrocities (14566); State Governments and Union Territories; District Administrations; Counsellors and Mental Health

Professionals; Law Enforcement Agencies; Rehabilitation and Welfare Authorities.

## 2. Proposed Solution

### 2.1 Overview and Core Idea

We propose a real-time, AI-assisted risk-triage layer that sits alongside the human helpline agent at every point of contact – voice call, IVRS, chatbot, mobile app, or web portal – and continuously analyses what a caller says and how they say it, surfacing a structured, multi-dimensional picture of their psychological state as the call happens. We are deliberately not building an “AI that detects emotions” or an “AI that diagnoses trauma.” We are building an AI-assisted real-time trauma-and-vulnerability triage layer that helps a helpline agent notice distress signals that could otherwise be missed during a high-pressure conversation. This assessment is distilled into a single, easy-to-read Stress Vulnerability Index (SVI) on a 0-100 scale, exactly as the problem statement asks, while a richer set of underlying risk signals – each carrying its own confidence estimate – drives every recommendation the agent sees.

The central design principle is this: the AI recommends; a trained human always decides. At no point does the system autonomously contact police, trigger witness protection, or take any irreversible action on its own. This is the core architectural choice, because in caste-atrocity cases the “obvious” escalation path can itself become a source of harm if the local police or administration is compromised, and because a wrongly triggered irreversible action can cause more damage to a victim than a missed one.

**Scope boundary, stated up front:** we detect conversational and acoustic indicators associated with elevated vulnerability; we do not diagnose depression, PTSD, dissociation, or any other psychiatric condition. Every dimension in Section 2.2 is an indicator surfaced for a trained human, never a clinical finding asserted by the system.

### 2.2 Eight Observable Risk Dimensions - Full Coverage of the Required Indicator List

Trauma is not one-dimensional, and a system that only tracks the two or three most dramatic symptoms will miss victims who present differently. To ensure every indicator named in the problem statement is explicitly and separately tracked – not folded silently into something else – we compute eight independent, evidence-based risk dimensions for every caller:

1. **Acute distress** – immediate emotional overwhelm (crying, panic, breathlessness)
2. **Depression indicators** – persistent low mood, hopelessness, anhedonia (“nothing matters now,” loss of interest in previously important things), flattened or monotone vocal affect, self-blame language – distinct from acute distress because it reflects a sustained state rather than a momentary spike

3. **Self-harm risk indicators** – linguistic and contextual signals potentially associated with suicidal ideation or self-harm intent, escalated for human assessment rather than treated as a diagnosis
4. **Fear of retaliation** – signals that the caller is afraid of future consequences from reporting (community backlash, loss of livelihood, family pressure to withdraw the complaint)
5. **Intimidation signals** – signals that the caller is currently being threatened or coerced – by the perpetrator, their associates, or local authority figures – distinct from fear of retaliation because it reflects an active, ongoing threat rather than an anticipated future one
6. **Dissociation** – signs that the caller has mentally disconnected from the conversation, often from severe trauma
7. **Social isolation** – indicators of being cut off from family, community, or support networks
8. **Chronic trauma indicators** – signs that this is not a first incident, but a pattern

Each dimension is scored independently and reported as a risk level plus a confidence estimate (e.g., “Fear of retaliation: High – 87% confidence”), so the agent always knows not just what the system found but how sure it is.

Mapping to the problem statement’s required indicator list, so every named term is traceably covered:

| Required Indicator    | Covered By                                                     |
|-----------------------|----------------------------------------------------------------|
| Severe trauma         | Acute distress + Chronic trauma indicators                     |
| Fear                  | Fear of retaliation                                            |
| Depression            | Depression indicators                                          |
| Suicidal ideation     | Self-harm risk indicators                                      |
| Intimidation          | Intimidation signals                                           |
| Social isolation      | Social isolation                                               |
| Extreme vulnerability | The overall SVI (Section 2.3) and Critical-risk categorization |

## 2.3 The Stress Vulnerability Index (SVI)

The SVI is computed as a documented, open, weighted combination of the eight dimensions:

$$\text{SVI} = w1.\text{acute\_distress} + w2.\text{depression} + w3.\text{self\_harm\_risk} \\ + w4.\text{fear\_of\_retaliation} + w5.\text{intimidation} + w6.\text{dissociation} \\ + w7.\text{social\_isolation} + w8.\text{chronic\_markers}$$

where  $w_i \geq 0$  and the weights sum to 1.

On the weights: for the prototype, the weights are configurable, evidence-informed, and expert-reviewable – not asserted as clinically validated. self\_harm\_risk and intimidation are given the highest starting weights, because missing genuine self-harm risk is the costliest possible error, and because active intimidation means the caller may be in immediate danger independent of what they are able to say on the call. In production, these starting weights would be reviewed and calibrated by qualified mental-health professionals and

domain authorities before deployment, and the weighting scheme itself – not just its output – is exposed to supervisors, not hidden inside the model.

This gives the Ministry, district administrators, and helpline supervisors exactly what they asked for – one number, on one scale, that instantly communicates urgency – while the actual clinical reasoning underneath remains multi-dimensional, transparent, and auditable. Nothing is hidden inside a black box: every SVI value can be traced back to exactly which of the eight dimensions drove it, and by how much (see Section 2.16, Explainable AI Panel).

## 2.4 Risk Categorization and What Happens at Each Level

| SVI Range | Category | What the Agent Sees and Can Do                                                                                                                                                                                                                                                                                                       |
|-----------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0-25      | Low      | <p>The AI runs quietly in the background, logging observations only. No action is suggested; the agent proceeds with the normal call flow.</p>                                                                                                                                                                                       |
| 26-50     | Moderate | <p>A passive risk panel becomes visible to the agent, showing all eight dimensions with confidence levels, without pushing any specific action.</p>                                                                                                                                                                                  |
| 51-75     | High     | <p>The system actively suggests appropriate support (counselling, legal aid, medical assistance) and asks the agent to confirm before anything is offered to the caller.</p>                                                                                                                                                         |
| 76-100    | Critical | <p>The system flags the call as requiring immediate attention. Any recommendation involving police intervention, witness protection, or non-medical emergency escalation is held behind a mandatory second layer of human sign-off from a senior reviewer before it can be acted on.</p> <p>Counselling, legal-aid, and medical-</p> |

assistance referrals can proceed faster, but only once the caller has re-confirmed, in the moment, that they want that specific help.

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## 2.5 How Recommendations Work - Automatic in Presentation, Human-Gated in Effect

The problem statement asks the system to “automatically recommend counselling, legal aid, medical assistance, police intervention, witness protection, or emergency support.” We satisfy this literally, across all six named support types, and we group them by how reversible each one is:

**Beneficial, low-risk referrals - counselling and legal aid.**

Support the caller can only benefit from, never be harmed by. Actioned quickly once the caller has been asked in the moment whether they want that support and the agent confirms.

**Medical assistance - split by urgency.** A non-urgent medical-assistance referral (e.g., connecting the caller to a health worker or clinic for after-the-fact care) follows the same fast, agent-confirmed path as counselling and legal aid. An urgent medical dispatch (e.g., the caller or a family member is currently injured and needs an ambulance) is fast-tracked for agent confirmation without requiring the senior-reviewer sign-off that gates police/witness-protection actions - because in a genuine medical emergency, the cost of delay can itself be life-threatening, and dispatching medical aid does not carry the same risk of institutional complicity that police involvement can carry in caste-atrocity cases. The distinction is deliberate: reversibility and risk-of-harm-from-the-response-itself, not just “how serious,” determines the approval path.

**Police intervention, witness protection, and non-medical emergency escalation.** Difficult or impossible to reverse, and in the specific context of caste-based atrocities, the very institution being invoked can sometimes be complicit in the harm. These require a second, distinct human - a senior reviewer - to explicitly approve before anything is triggered, enforced as a hard rule in the decision logic so the action can never reach an executed state without that sign-off having actually occurred.

## 2.6 Analysing Voice: What We Listen For, Beyond Words

Most systems that attempt to detect distress from speech rely on basic sentiment analysis of words alone, or at best a shallow read of tone and pitch. In line with the problem statement’s call for NLP, Speech Analytics, and Emotion AI, we combine all three as separate, complementary evidence layers - while being explicit that none of

them, alone or combined, lets the system determine with certainty how a caller feels. They are supporting evidence for a trained human, not a verdict.

**NLP / lexical content** – what the caller actually says, analysed for direct and indirect disclosures of distress, fear, depression, intimidation, isolation, or self-harm risk, in Hindi, English, and naturally code-switched Hindi-English speech, without translating the caller’s words into English first.

**Speech Analytics / acoustic features** – pitch variation, jitter, shimmer, energy, and speaking rate, extracted directly from the raw audio.

**Emotion AI / paralinguistic modelling** – industry-standard affect-recognition modelling applied to the acoustic layer (e.g., flattened affect as a depression indicator, tremor or breathlessness as acute-distress indicators). We use this as a supporting signal to the lexical content, not as the primary decision-maker, because acoustic-only emotion detection is known to be unreliable across different speakers, dialects, and recording conditions – a limitation we surface to the agent via the confidence score (Section 2.17), not one we paper over.

**Silence itself, treated as a signal** – the component we consider most distinctive about our approach. Nearly every voice-analysis system treats a pause as noise to be filtered out. We instead treat silence as a structured, meaningful signal: unusual silence is treated as an observable signal that may contribute to the dissociation and acute-distress scores, not as proof that dissociation is occurring. Severe dissociation can present as speech simply stopping mid-sentence for several seconds before resuming as if nothing happened, so an unexplained, grammatically unmotivated pause is scored as evidence worth surfacing to the agent – one input among several, never a standalone diagnosis. The system measures how long a silence lasts, precisely where it falls in the sentence, what word came immediately before it, and whether the pause overlaps with signs of crying or irregular breathing.

## 2.7 Multilingual Design

Support for Indian languages and dialects is a first-class design requirement, not an add-on. Rather than translating every call into English and running a single English-language model, the system uses language-native processing that preserves the caller’s own words, dialect, and natural code-switching between languages, so idioms of distress specific to a region or community are recognised as they are actually spoken. This design choice is directly supported by current research into cross-lingual mental-health language processing for Indian languages, which documents that expressions such as “man ka bhoj” (a burden on the mind) carry culturally specific meaning entirely absent from Western clinical vocabularies and standard NLP tokenizers, and that code-mixed Hindi-English phrasing measurably reduces the accuracy of systems that are not built to handle it directly.

Published research on speech-based distress detection also shows that pause and silence patterns are not uniformly informative across every language – their predictive strength varies. We treat this as a design constraint rather than an assumption: the Silence Fingerprint’s contribution to each risk dimension is calibrated and validated separately for each supported language, and if a given language

shows the silence signal is not reliably informative for that language's speech patterns, the system relies more heavily on the lexical and acoustic signals for that language rather than assuming a one-size-fits-all silence model transfers cleanly across Hindi, English, and code-switched speech.

## 2.8 Consent, Privacy, and Confidentiality

Because this system listens to and analyses some of the most sensitive disclosures a person can make, we treat consent, privacy, and confidentiality as three related but distinct commitments.

Consent is an ongoing, revocable relationship with the caller, not a one-time checkbox. Three separate permissions are tracked independently: whether the call may be recorded at all, whether the AI may analyse it, and whether the anonymised data may later be used to improve the system. Consent is re-checked during the call itself, at moments the system detects as significant, and this re-check is initiated by the system, not the human agent, so the emotional dynamic between caller and agent is never used to extract consent. If the caller revokes consent at any point, AI analysis stops immediately, while the human agent's ability to help the caller continues uninterrupted.

Confidentiality is enforced as a set of concrete controls, not just a policy statement:

- **Encryption** of call recordings and transcripts at rest and in transit.
- **Need-to-know access control** – a case record is visible only to the agent who handled the call and the specific supervisor assigned to that case; there is no general cross-case search or browsing of victim records.
- **Audit logging** – every access to a caller's recording, transcript, or risk profile is logged with who accessed it and when, so any inappropriate access is independently discoverable after the fact.
- **Defined retention and deletion** – recordings and transcripts are retained only as long as the case requires, with a caller-initiated deletion request honoured for any data covered by the consent they gave.
- **No re-identification in aggregate views** – the administrator analytics dashboard (Section 2.19) shows only anonymised, aggregate statistics; individual victim identities are structurally excluded from that view, not merely hidden by convention.

We take this last point further than a general assurance, because it is precisely where a comparable real-world system failed. In 2022, a major crisis-text helpline was found to have shared conversation data – described by the organisation as “anonymised” – with a for-profit commercial partner, and privacy researchers subsequently noted that data described as anonymised in other contexts has repeatedly been shown to be re-identifiable. Separately, the helpline's own data-ethics advisory board was not consulted before the decision was made, despite existing on paper for that exact purpose. We treat both of these as specific, named failure modes to design against, not general cautionary tales. First, on anonymisation: aggregate dashboard figures are subject to a minimum group-size threshold before being displayed at all (a statistic covering fewer than a defined minimum number of cases is suppressed rather than shown), precisely to

prevent a small, identifiable group – for instance, a specific village or a specific rare risk combination in a specific district – from being re-identifiable even in “aggregate” form. Second, on oversight: any proposal to use caller data beyond the direct purpose of helping that specific caller – including the opt-in “improve the system” pathway described above – requires active, minuted sign-off from a standing data-ethics review body that meets on a fixed schedule regardless of whether a proposal is pending, specifically so that oversight cannot quietly lapse the way it did in the comparable real-world case. No secondary use of caller data proceeds on the strength of a terms-of-service disclosure alone.

## **2.9 Human Oversight as a Layered Safety Net, Not a Single Checkpoint**

Three independent layers exist so a single missed signal – by the AI or by a person – does not go unnoticed: (1) the agent handling the call in real time, who retains full authority to override, modify, or ignore any AI suggestion; (2) a scheduled follow-up at 72 hours and again at 7 days for any caller assessed as Low or Moderate risk; and (3) supervisory review after the fact, where a portion of Low/Moderate calls are spot-checked and every Critical-risk call is reviewed the same day.

## **2.10 What the System Explicitly Does Not Do**

We detect conversational and acoustic indicators associated with elevated vulnerability; we do not diagnose depression, PTSD, dissociation, or any other psychiatric condition. The system does not claim to determine with certainty what a caller is feeling. It detects observable, describable signals – specific words, specific patterns of silence, specific vocal characteristics – and presents them to a trained human as decision-support information, each paired with a confidence estimate. Its job is to make sure a distress signal is never missed by an overworked or momentarily distracted human agent, not to replace the clinical and legal judgment that only a trained person can exercise. This applies equally to all eight dimensions: every dimension name (including “Depression indicators” and “Dissociation”) describes a pattern the system flags for human review, not a clinical finding the system asserts.

## **2.11 Degradation Under Real-World Conditions**

The system is designed to fail safely rather than fail silently. If speech recognition becomes unavailable mid-call, the system falls back to keypad-based IVRS input and tells the agent explicitly that no transcript is available. If voice analysis fails, the system continues scoring based on transcript content and the agent’s own notes. If confidence across all signals drops too low to be reliable, a visible “Low confidence” uncertainty indicator is raised, the threshold for suggesting any high-stakes action is deliberately raised, and the call is routed to a human reviewer rather than the system guessing. If the connection drops entirely, no captured data is silently discarded, and

processing resumes cleanly on reconnection. In every failure scenario: the system never takes or suggests an irreversible external action while it is uncertain.

A distinct and, in practice, more common risk than an outright component failure is silent accuracy drift: a system that keeps running and keeps producing scores, but whose scores quietly become less reliable as real-world conditions – background noise, an unfamiliar dialect, a poor-quality rural line – diverge from the conditions it was calibrated on. Published research on multimodal distress-detection pipelines specifically warns that this kind of fusion model degradation under real-world distribution shift is common and easy to miss, precisely because nothing crashes and no error is raised. We monitor for this directly rather than assuming it away: the rate at which agents override or contradict the system’s suggested actions is tracked on an ongoing basis as a proxy signal, and a sustained rise in override rate for a specific language, dialect, or call-quality condition triggers the same fairness-style review described in Section 2.14, even though nothing in the pipeline has technically “failed.”

## 2.12 Dataset and Validation Strategy - Honesty About What Powers the Model

There is currently no large, publicly available Indian dataset of caste-atrocity-related trauma disclosures, and we do not pretend otherwise. We treat this as the solution’s single biggest open risk and address it directly rather than glossing over it, in three parts.

**What powers the prototype today.** The system’s word-and-phrase-level detection is built to be transparent and directly traceable – every score can be explained by pointing to the exact words or silence pattern that caused it, rather than relying on an opaque model trained on data that does not exist. Where deeper language understanding is used to catch indirect disclosures, negation, or unusual phrasing, it is calibrated using a small set of realistic vignettes developed together with licensed mental-health professionals, used strictly to fine-tune and calibrate, not to replace the transparent, rules-based foundation.

**How we validate what we have, honestly.** Rather than an unverifiable accuracy claim, the prototype is evaluated against an expert-labelled vignette test set held out from calibration, and we report the metrics that actually matter for a safety-critical triage tool: recall on the highest-severity vignettes (did the system ever miss a case an expert flagged as high-risk?), false-negative rate specifically on self-harm and intimidation vignettes (the two dimensions where a miss is most costly), and inter-rater agreement between the system’s dimension-level output and the mental-health professionals’ independent labelling of the same vignette. We report these numbers as they are, including where they are weak, rather than presenting a single polished accuracy figure.

To make this concrete rather than aspirational, we anchor our target against the published performance of comparable, already-deployed crisis-line machine-learning systems: a recent peer-reviewed evaluation of a real crisis-text-line suicidality classifier reported a false-negative rate near 17% and a false-positive rate near 25% in production use. Our stated goal for the self-harm-risk and intimidation dimensions – the two where a miss is most dangerous – is to meet or

improve on that published false-negative baseline during pilot validation, and we treat falling short of it as a reason to delay wider deployment of that specific dimension, not a footnote to explain away.

**How we close the data gap over time, without waiting for it to be closed to be useful.** The system is deliberately architected so that (a) it is useful and safe on day one using the transparent, rules-plus-vignette foundation, and (b) it improves in a structured way as real data becomes available, without ever requiring an architecture change. The path to real data is itself consent-gated and ethics-reviewed: only calls where a caller has explicitly opted in to “anonymised data may be used to improve the system” (Section 2.8) ever become training signal, aggregation and anonymisation happen before any such data leaves the helpline’s systems, and mental-health professionals remain in the loop to review and label before any retraining. Until that pipeline has produced a dataset large enough to meaningfully change the model’s behaviour, the safety architecture is the actual mitigation for the data gap: low-confidence outputs are visibly flagged (Section 2.17) and are structurally prevented from triggering high-stakes actions (Section 2.5, Section 2.11) regardless of how the underlying scores were produced. In other words, we do not need the dataset to exist yet for the system to be safe to deploy – we need the human-gating and confidence-reporting to be strict, which they already are.

## 2.13 Measured Real-World Performance

We report measured, not estimated, processing speed end-to-end on standard, non-specialized hardware (CPU-only, no dedicated GPU):

| Processing Stage               | Measured Time                                                  |
|--------------------------------|----------------------------------------------------------------|
| Speech-to-text transcription   | ~4.15 s per 2-second audio window (mean of 20 runs, SD 0.12 s) |
| Silence-pattern detection      | under 1 ms                                                     |
| Eight-dimension risk scoring   | under 1 ms                                                     |
| SVI calculation                | under 1 ms                                                     |
| Suggested-action determination | under 1 ms                                                     |

We are direct about the implication of this table: on unoptimized CPU-only hardware, the speech-to-text stage alone runs slower than real time, and this is currently the bottleneck of the whole pipeline. Our answer to a judge asking “how is 4.15 seconds real-time?” is: the architecture is streaming, and the current CPU-only implementation establishes functional correctness end-to-end – it demonstrates that the pipeline works, not the latency of a production deployment. All downstream risk-scoring stages already execute in under 1 millisecond. Production deployment would use a distilled or otherwise optimized speech-to-text model, or GPU/edge acceleration, either of which reliably brings this stage down to a fraction of a second. We report the unoptimized number rather than a projected one because an honest, verifiable measurement is worth more to a technical judge than an optimistic estimate we haven’t yet demonstrated.

## 2.14 Fairness Across Callers

The system's accuracy at correctly identifying genuine risk is checked separately across caller gender, language and dialect, and recording quality. If the system performs meaningfully worse for any one group than another on any risk dimension, that specific dimension is not used in production for that group until the gap is closed. This is a hard constraint on deployment, not an aspirational goal.

## **2.15 Why This Approach Is Better Than the Common Alternative**

The conventional approach treats this as an emotion-detection problem: analyse tone, run a sentiment classifier, produce a single stress score, and pass it to a rule engine. Our solution differs in three ways. First, we treat this explicitly as a triage problem borrowed from the same logic that governs emergency-room and disaster medicine. Second, we refuse to let a single number make a high-stakes decision by itself, preserving the full nuance of an eight-dimension, confidence-scored assessment underneath a single headline figure. Third, we treat human judgment as the irreplaceable final layer for anything irreversible, which directly protects against the single greatest risk in this domain: a system that is confidently wrong at exactly the moment a vulnerable person's safety depends on it being right.

## **2.16 Explainable AI Panel**

The agent screen never shows a bare number. Alongside the headline SVI (e.g., "Critical - 78/100"), a breakdown panel shows each of the eight dimensions with a colour-coded level (e.g., Acute distress: High; Depression indicators: Moderate; Self-harm risk indicators: Moderate; Fear of retaliation: High; Intimidation signals: High; Dissociation: Moderate; Social isolation: High; Chronic trauma: Moderate), plus a plain-language "why" line generated directly from the specific signals that drove the score - for example, "Long mid-sentence pause detected," "Statement indicating fear of retaliation," "Flattened vocal affect consistent with depression indicators." This turns the SVI from a number the agent has to trust into a number the agent can inspect.

## **2.17 Confidence and Uncertainty Reporting**

Every one of the eight risk dimensions is reported as risk level + confidence, not risk level alone (e.g., "Intimidation signals: High - 87% confidence," "Depression indicators: Moderate - 72% confidence"). A separate, system-wide AI Confidence indicator sits above the eight-dimension panel (e.g., "AI Confidence: 91%," or "Low confidence - noisy audio"), and whenever this drops below a safe threshold the interface explicitly states that the AI recommendation should not be used for high-stakes action on its own.

Current research on clinical decision-support interfaces shows that displaying uncertainty is generally valuable for trust and appropriate reliance, but also that a dense, statistically detailed presentation of that uncertainty can itself raise a user's stress at exactly the moment they most need to think clearly. We take this finding seriously in the interface design rather than only in the underlying model: the

confidence indicator shown to the agent is deliberately simple – a single percentage and a colour, not a breakdown of the statistical method behind it – so that communicating honest uncertainty does not become its own source of pressure on an already-stretched helpline agent.

## 2.18 Case Timeline

Alongside the live risk panel, the agent screen keeps a running, timestamped timeline of the call, e.g.:

```
00:12 -- Distress detected
00:24 -- Intimidation signal detected
00:37 -- Long unexplained silence detected
00:48 -- SVI increased 42 -> 67
01:02 -- High-risk recommendation generated
```

This gives the agent – and any supervisor reviewing the call afterward – a reconstructible narrative of exactly when and why the assessment changed, supporting the same-day review of every Critical-risk call described in Section 2.9.

## 2.19 Administrator Analytics Dashboard

Beyond the individual caller, a district/state-level dashboard presents anonymised, aggregate information for supervisors and administrators: number of cases, risk-level distribution, language distribution, average response time, counselling / legal-aid / medical-assistance referral volumes, high-risk case counts, follow-up completion rates for the 72-hour/7-day check-ins, and ongoing false-positive/false-negative monitoring to support the fairness commitment in Section 2.14. Individual victim identities are never exposed in this aggregate view, consistent with the confidentiality controls in Section 2.8.

## 2.20 The Silent SOS – A Duress Signal for When Speaking Freely Is Not Safe

Every capability described so far assumes the caller is free to speak honestly about what is happening to them. In a meaningful number of real caste-atrocity cases, this assumption does not hold: a victim may be calling from inside the home of the person threatening them, in the presence of family members who told them not to call, or under direct observation by someone who would retaliate if the call's true purpose were discovered. No system that relies only on what a caller says or how they sound can help someone who cannot safely say or sound anything unusual at all. This is not a gap named anywhere in the problem statement, and it is not a gap addressed by any comparable helpline or crisis-line system we are aware of – which is precisely why we consider it worth solving.

The mechanism is a pre-agreed, silent duress code, conceptually similar to a bank's duress PIN. At the start of any call or IVRS session, alongside the ordinary consent flow, the caller is told – once, and only if they choose to engage with this feature – that they may at any point enter a specific keypad sequence silently, without saying anything out

loud, to indicate that they are not able to speak freely. Entering this code does not change anything audible on the caller's end: the call continues to sound completely ordinary to anyone who might be listening nearby. What changes is entirely on the system and agent side. The agent's screen immediately and discreetly indicates that a duress signal has been received, and the agent's script silently shifts to a mode built for exactly this situation: closed, yes-or-no questions that the caller can answer with a single keypress rather than a spoken sentence ("If you are in the same location as the person who harmed you, press 1. If you are safe to continue speaking normally in a few minutes, press 2."). The SVI and risk dimensions continue to update based on whatever signal is available, and the case is automatically flagged for the highest tier of urgency and same-day supervisory review, regardless of what the audible portion of the call did or did not contain.

This feature is included because it directly serves the single most vulnerable subset of the population this problem statement is about: victims who are being actively controlled or surveilled by the person who harmed them at the exact moment they are trying to seek help. It requires no exotic technology – keypad-tone detection during a live call is a well-established, decades-old capability, and silent duress signalling itself is a long-established security pattern used in banking and physical-security contexts, and, closer to this domain, at least one domestic-violence hotline has reportedly used a comparable silent keypad signal. What has not been done, as far as we can find, is integrating this signal into a continuously-updating, AI-assisted trauma-triage pipeline, where the signal doesn't just trigger a single alert but dynamically reshapes the entire remainder of the interaction.

We also design specifically against the reason similar mechanisms have failed elsewhere. A U.S. Federal Trade Commission review of duress-PIN mechanisms at ATMs found they were never meaningfully adopted in practice, concluding that the operational cost and the risk of false alarms outweighed the benefit. We address both failure modes directly. False alarms are reduced by requiring a specific, multi-digit sequence deliberately chosen to be extremely unlikely to be entered by accident during ordinary call navigation, rather than a single keypress. And unlike a mechanism that triggers an irreversible external dispatch on its own, a Silent SOS signal here does not bypass the human-gating architecture described throughout this document – it raises the call's priority and reshapes the agent's script, but any subsequent police or witness-protection action still passes through the same senior-approval gate as every other Critical-risk recommendation (Section 2.5), so a false trigger costs a more cautious conversation, not an unwarranted dispatch.

## **2.21 Tamper-Evident Audit Trail for Legal and Evidentiary Integrity**

The audit log described throughout this document (Section 2.8) already records every access to a caller's data. We extend this into a tamper-evident record: each audit-log entry includes a cryptographic hash of the entry immediately before it, forming a hash chain across the entire log for a given case. If any entry in the chain were ever altered or deleted after the fact, the hash chain from that point forward would no longer verify – making tampering mathematically detectable rather than merely against policy.

This matters specifically in this domain because a victim's first disclosure, and the system's contemporaneous, timestamped assessment of their distress and vulnerability at that moment, may later become relevant to legal proceedings under the SC/ST (Prevention of Atrocities) Act – for instance, in establishing that a credible complaint and a clear risk signal existed at first contact, independent of what happened administratively afterward. A tamper-evident log means a court, a review body, or the caller's own legal counsel can trust that the record of what the system detected, and when, has not been quietly altered by anyone with system access after the fact. This is a comparatively small addition to what has already been built, and it converts the audit log from an internal operations tool into something that can credibly support the victim's own legal position later.

## **2.22 Closing the Referral Black Hole - Real-Time Availability and Warm Hand-off**

The most common point of failure in victim support systems generally is not detection – it is what happens immediately after a referral is made. A victim told to “please contact this counsellor” or handed a phone number for legal aid very often never follows up, not from lack of need but because the moment of willingness passes, the number goes unanswered, or the service is at capacity. This failure mode is well documented across support services generally and is not specific to this helpline, which is exactly why solving it here is a meaningful, transferable improvement rather than a narrow fix.

When a counselling or legal-aid referral is confirmed during a call (Section 2.5), the system checks real-time appointment availability at the nearest appropriate, already-empanelled counselling centre or legal-aid clinic, and offers to book a specific slot immediately, live, while the caller is still on the line – a warm hand-off with a confirmed appointment, not a name and a number to follow up on alone later. Where the caller consents, the receiving counsellor or legal-aid worker is given the case's risk-dimension summary in advance (Section 2.16), so the victim does not have to re-explain their situation from nothing to a second stranger immediately after having just disclosed it to the first.

This directly strengthens the “who benefits and how do we measure it” question central to demonstrating real impact: the system does not stop at generating a recommendation, but measurably improves the odds that the recommended support is actually received, which is the outcome that was always the point.

## **2.23 Agent Secondary-Trauma Protection - Protecting the Protector**

Every feature described so far protects the caller. This one addresses a real and well-documented problem that the problem statement does not ask about at all: helpline agents who spend entire shifts absorbing repeated, severe distress disclosures are themselves at serious risk of vicarious trauma and burnout, a documented occupational hazard across crisis-line and helpline staffing generally. An agent who is

quietly burning out is also a less reliable safety net for the very victims this system exists to protect, which makes this a genuine extension of the solution's own goals, not a tangential addition.

We propose applying the same acoustic and conversational analysis techniques already built for callers (Section 2.6), pointed instead – with the agent's own informed consent, as a workplace-wellbeing tool they opt into, never a covert performance-monitoring system – at the agent's own voice across the course of a shift. Signs of accumulating strain (flattening vocal affect, slowing response times, increasingly short or clipped responses across successive calls) generate a private, supervisor-visible signal suggesting the agent may benefit from a break, never a score attached to their name for evaluation purposes, and never something the agent themselves cannot see or dispute. The framing is deliberately protective, not evaluative: this exists to catch an agent quietly struggling before it affects their wellbeing or a caller's experience, in the same spirit as the victim-facing safety net described in Section 2.9.

Existing crisis-line practice for detecting vicarious trauma and burnout relies almost entirely on periodic self-report check-ins and clinical supervision, and published reviews of this practice specifically note that most studies in this area do not use any validated measurement instrument at all. Rather than inventing our own ad hoc heuristic for what counts as “strain,” we anchor the acoustic signal against an already-validated, widely used occupational instrument for exactly this purpose – the Professional Quality of Life Scale (ProQOL), which specifically measures compassion fatigue and burnout in helping professions – so the automated signal is calibrated against a real clinical standard rather than a number we invented, and so its output can be meaningfully compared against the periodic self-report the agent would otherwise fill out.

This reframes the system as a whole from “a tool that watches victims” to “a system that safeguards everyone in the chain of care” – and we are not aware of this being addressed by any comparable helpline technology, which is precisely the kind of unclaimed, genuinely useful ground worth occupying.

## **2.24 Accessibility Bridge for Non-Verbal Disclosure**

Everything described so far is built around a voice call. A meaningful population the problem statement's own list of channels already gestures toward – via its inclusion of the chatbot, mobile application, and portal – cannot or should not rely on the voice channel at all: callers who are deaf or hard of hearing, callers with speech-affecting disabilities, and callers for whom typing quietly is the only safe way to communicate in their present circumstances.

We extend the same eight-dimension risk analysis (Section 2.2) to the text-based channels using exactly the lexical layer already built for voice calls (Section 2.6), so a caller interacting purely through the chatbot or SMS-based portal receives the identical quality of trauma-and-vulnerability assessment as a caller on a voice line – the same SVI, the same explainable panel, the same human-gated recommendations – rather than a stripped-down, second-class version of the system. This is a comparatively small engineering extension, since the lexical analysis layer is channel-agnostic by design, but it

ensures that a population already explicitly included in the problem statement's list of approved interfaces is not, in practice, given a materially inferior version of the protection this system provides.

## 2.25 The 36-Hour Build Plan - MVP First, Wow-Factors as Ordered Stretch Goals

This build plan deliberately separates two things that are easy to blur under time pressure: the MVP (Section 2.29) that must be complete and stable on its own, and every enhancement layered on top of it. The sequencing below builds and hardens the MVP first, in full, before a single hour is spent on any stretch-goal feature – because a working core that honestly satisfies the problem statement beats an impressive-looking pile of half-finished features every time a judge asks to see something actually run.

**Hours 0-3 - Foundation.** Finalize technology choices, assign ownership of each MVP pipeline stage, and lock the exact data format passed between components before writing code.

**Hours 3-12 - MVP core pipeline, built and tested in isolation.** Speech-to-text with word-level timing; a simple duration-based silence flag (not yet the full placement-classified Silence Fingerprint); the eight-dimension rules-layer risk scorer; SVI and bucket calculation – each built and verified independently against controlled test inputs.

**Hours 12-20 - MVP end-to-end wiring.** Connect the verified components into a single live pipeline with a real call-style interface, live-updating risk scores and SVI, a basic suggested-actions display, the human-in-the-loop confirm/override layer, and – built and tested with the same priority as everything else in this window, never deferred – the senior-approval gate for police/witness-protection/emergency actions and a single, clear, upfront consent step.

**Hours 20-24 - MVP integration testing.** Test with messy, real-world-like input: pauses in unexpected places, Hindi-English code-switching, background noise, a caller who barely speaks. By the end of hour 24, the MVP must stand alone as a complete, demoable system that fully and literally satisfies the Problem Statement's Expected Solution Requirements (Section 1.3) – with zero stretch-goal features attempted yet.

**Hours 24-36 - Stretch goals, attempted strictly in this priority order, each begun only once the previous one is fully working:**

1. **(Hours 24-27) The Silent SOS (Section 2.20).** Highest-priority stretch goal, because it protects the single most vulnerable subgroup this problem statement is about.
2. **(Hours 27-29) Agent Secondary-Trauma Protection (Section 2.23).** Second priority – the most genuinely novel addition, and self-contained enough to build and test quickly on top of the existing acoustic layer.
3. **(Hours 29-30) Tamper-Evident hash-chained audit log (Section 2.21).** A cheap, low-risk upgrade of the MVP's existing plain audit log.
4. **(Hours 30-33) Polish layer.** The full Silence Fingerprint (placement classification, crying/breathing overlap), the Explainable “why” panel, and per-dimension confidence reporting.

5. **(Hours 33-35) Referral real-time availability and warm hand-off (Section 2.22)**, demonstrated against a small, mocked directory of counselling/legal-aid slots.
6. **(Hours 35-36) Rehearsal**. The Case Timeline, Administrator Analytics Dashboard, and a dedicated, polished Accessibility Bridge experience (Section 2.24) are presented as documented roadmap items on slides if not reached, rather than attempted live at the last hour.

If time runs out at any point in this sequence, the team stops and demos the MVP plus whichever stretch goals are fully complete – never a partially wired stretch feature sitting on top of an otherwise-solid core.

## 2.26 What the Prototype Looks Like

The finished prototype is two connected screens, run on two machines side by side, mirroring a real helpline interaction.

**The caller-facing screen** resembles a simple, phone-style interface: a caller-ID-style header identifying the line as NHAA 14566, a language selector for Hindi, English, or mixed Hindi-English, a call-connect button, an active call timer, and a live waveform. Before the call is treated as connected, a short, plainly worded consent message is shown and read aloud, and the caller’s choices are recorded before anything else happens. A brief, optional mention of the Silent SOS keypad code (Section 2.20) is included in this same consent step, worded carefully so its existence is available to anyone who needs it without drawing attention to it.

**The agent-facing screen** is where the system’s reasoning becomes visible. A live transcript builds line by line in the language actually spoken, without forced English translation. Beneath it, eight horizontal bars represent the eight risk dimensions, each filling and changing colour in real time, paired with the confidence indicator from Section 2.17. Above these, the headline SVI is shown as a large, colour-coded figure – green/yellow/orange/red for Low/Moderate/High/Critical. A distinct visual alert appears when the system detects an unusual, unexplained silence, and a separate, unmistakable alert appears if a Silent SOS code is entered, silently switching the on-screen script to the yes/no duress format. The explainable “why” panel (Section 2.16) and the running case timeline (Section 2.18) sit alongside the risk bars. Below all of this sits the suggested-actions panel: counselling, legal-aid, and medical-assistance options the agent can confirm quickly once the caller agrees – with a live, bookable appointment slot shown directly in the panel (Section 2.22) – a fast-tracked emergency-medical-dispatch option for acute physical danger, and police-intervention or witness-protection options that remain visibly locked, marked as requiring a second, senior approval, until that sign-off actually happens.

A separate administrator view (Section 2.19) demonstrates the aggregate dashboard for judges evaluating scalability beyond a single call, and a small supervisor-only panel demonstrates the agent-wellbeing signal (Section 2.23) alongside it, framed explicitly as a wellbeing safeguard rather than a performance metric.

Together, these screens let anyone watching the prototype see the entire intended interaction unfold exactly as designed: a caller speaks naturally in their own language – or signals silently if they cannot

speak freely at all – the system listens and reasons across text, voice, and silence simultaneously across all eight required indicators, the risk picture builds visibly and honestly in front of the agent, every referral is followed all the way through to a real, booked outcome rather than stopping at a recommendation, and any action serious enough to be irreversible remains visibly, provably gated behind a human decision at every step – AI recommends; trained humans decide.

## 2.27 How This Solution Compares Against Documented Limitations of Existing Systems

Before finalising this design, we reviewed the published literature on comparable systems – speech-based depression and PTSD detection, crisis-hotline machine learning, duress-signalling mechanisms, and vicarious-trauma monitoring – specifically to identify the limitations those systems are documented to have, rather than assuming our own design was already complete. The table below is that review, stated plainly, including the points where our first design fell short and has since been revised.

| # | Documented Limitation in Existing Research or Deployed Systems                                                           | Addressed in This Solution                                                                                                                         |
|---|--------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Cross-lingual/cross-dataset models degrade sharply outside training conditions                                           | Yes – transparent rules-layer detection is language-agnostic by design, not dependent on cross-lingual transfer of a trained model                 |
| 2 | Training datasets in this space are small and non-representative (e.g., 189 subjects in the standard academic benchmark) | Yes – explicitly acknowledged; the system uses calibration on a small expert-built vignette set, not full training, and states this limit honestly |
| 3 | Acoustic-only emotion detection is unreliable across speakers, dialects, and recording conditions                        | Yes – acoustic signals are explicitly treated as supporting evidence only, never the primary decision-maker                                        |
| 4 | Pause/silence signal strength is not consistent across all languages                                                     | Yes – the Silence Fingerprint is calibrated and validated per language, with reduced reliance on it for any language where it proves               |

|    |                                                                                                             |                                                                                                                                                                                           |
|----|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |                                                                                                             | unreliable                                                                                                                                                                                |
| 5  | Even the best published crisis-line machine-learning systems retain meaningful error rates                  | Yes – a concrete numeric target (matching or beating a published false-negative baseline) is stated for the highest-stakes dimensions, rather than an unverifiable claim of high accuracy |
| 6  | Model training can be unstable and introduce class-specific bias depending on random initialization         | Yes – the deterministic rules layer is inherently stable; any optional learned layer is subject to explicit stability testing before deployment                                           |
| 7  | Most deployed crisis-line systems lack explainability for their risk scores                                 | Yes – exceeded, via the explainable panel showing the specific signals behind every score                                                                                                 |
| 8  | Demographic bias frequently goes undetected in production systems                                           | Yes – fairness auditing is a hard deployment gate, not an aspirational goal                                                                                                               |
| 9  | A one-time terms-of-service click does not constitute meaningful consent from someone in crisis             | Yes – exceeded, via granular, revocable, system-initiated ongoing consent                                                                                                                 |
| 10 | Data described as “anonymised” has repeatedly been shown to be re-identifiable in practice                  | Yes – minimum group-size thresholds are enforced before any aggregate statistic is displayed, specifically to prevent small-group re-identification                                       |
| 11 | Ethics oversight bodies have been bypassed in real data-use decisions at comparable organisations           | Yes – any secondary use of caller data requires active, minuted sign-off from a standing, regularly-meeting ethics body, not a dormant one                                                |
| 12 | Duress-signal mechanisms have historically been abandoned due to false-alarm rates and cost-benefit failure | Yes – a deliberately low-false-trigger-probability sequence is required, and any resulting escalation still passes through the same human-approval gate as any                            |

|    |                                                                                                            |                                                                                                                                                 |
|----|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
|    |                                                                                                            | other recommendation                                                                                                                            |
| 13 | Displaying AI uncertainty can itself increase user stress if presented in a dense or technical form        | Yes – the confidence indicator shown to agents is deliberately simple (a percentage and a colour), not a statistical breakdown                  |
| 14 | Vicarious-trauma monitoring in crisis workers typically relies on self-report with no validated instrument | Yes – the agent-wellbeing signal is calibrated against the ProQOL scale, a validated occupational instrument, rather than an invented heuristic |
| 15 | Multimodal pipelines can silently degrade in real-world accuracy without any visible failure               | Yes – ongoing monitoring of agent override rates by language, dialect, and call quality is used as a proxy signal for this kind of silent drift |

Every limitation we could find documented against a comparable real system or in the relevant published research is addressed above, with the specific design change made and the specific source that motivated it. Where an existing solution’s flaw was structural (for example, consent-by-click-through, or an ethics board that exists without real authority), our response is also structural, not a policy statement layered on top of an unchanged design.

## 2.28 Existing Solutions Today, Compared Directly Against This Solution

The table below compares this solution against every category of real, currently operating or published system relevant to this problem: international and Indian crisis helplines, published academic distress-detection research, and established physical-security duress mechanisms. Assessments of external systems are based only on what each organisation has publicly documented or what has been independently published about them, since none of these systems’ full internal implementations are publicly disclosed; where something is not publicly documented, it is marked as such rather than assumed absent.

| Capability                                           | Crisis Text Line (US)       | iCall (India)                   | Vandrevala Foundation (India)   | S.           |
|------------------------------------------------------|-----------------------------|---------------------------------|---------------------------------|--------------|
| Tracks multiple independent risk dimensions, not one | Partial – ML risk flag, not | No – human counsellor judgement | No – human counsellor judgement | N<br>c<br>ju |

|                                                                                                                      |                                                          |                                                      |                                                      |                                                      |
|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|------------------------------------------------------|------------------------------------------------------|------------------------------------------------------|
| score                                                                                                                | multi-axis                                               | only                                                 | only                                                 | or                                                   |
| Explainable “why” evidence shown to the person acting on it                                                          | Not publicly documented at deployment                    | Not applicable (human judgement)                     | Not applicable (human judgement)                     | Not applicable (human judgement)                     |
| Reports confidence/uncertainty alongside every signal                                                                | Not publicly documented                                  | Not applicable                                       | Not applicable                                       | Not applicable                                       |
| Native multilingual, code-switching-aware (no translation step)                                                      | English-primary                                          | Yes – multiple Indian languages, human-mediated      | Yes – multiple Indian languages, human-mediated      | English-primary                                      |
| Treats silence/pauses as a structured signal                                                                         | No                                                       | Not applicable (human intuition only)                | Not applicable (human intuition only)                | Not applicable (human intuition only)                |
| Human-gate for irreversible actions enforced in code, not policy                                                     | Not publicly documented                                  | Yes, but as trained-human process, not code-enforced | Yes, but as trained-human process, not code-enforced | Yes, but as trained-human process, not code-enforced |
| Silent duress/SOS signal for a caller who cannot speak freely                                                        | No                                                       | Not publicly documented                              | Not publicly documented                              | Not publicly documented                              |
| Standing, actively-convened ethics oversight of data use                                                             | Documented failure – board existed but was bypassed      | Not publicly documented                              | Not publicly documented                              | Not publicly documented                              |
| Monitors and protects the wellbeing of the human responder                                                           | Relies on self-report; documented gap in validated tools | Relies on supervision and self-report                | Relies on supervision and self-report                | Relies on supervision and self-report                |
| Designed specifically around caste-atrocity context (fear of a compromised authority, intimidation, chronic pattern) | No – general crisis population                           | No – general mental-health population                | No – general mental-health population                | No – general mental-health population                |

The pattern across this table is consistent: existing crisis-line services are strong on trained human empathy but structurally lack multi-dimensional, explainable, confidence-scored AI assistance; existing academic research systems have the beginnings of the underlying detection techniques but are not built as deployable, human-gated, consent-governed operational systems; and existing physical-security duress mechanisms solve the “cannot speak freely” problem in isolation but were never integrated into an ongoing psychological assessment. No system we could identify combines all of these into one design – which is the specific gap this solution is built to fill.

## 2.29 MVP Scope - The Core This Solution Cannot Work Without

Everything described in this document up to this point is the complete solution. This section exists to answer a different, equally important question: if time, resources, or a 36-hour clock forced a cut, what is the smallest version of this system that still honestly and safely does what the problem statement asks – and what is genuinely an enhancement layered on top of that core, not a corner cut from it?

The guiding principle for this MVP is that it must be noble by construction: every feature removed from it is removed because it refines something the MVP already does safely, never because it removes a safeguard. Two things are never deferred, at any build stage, no matter how little time remains: the human-approval gate for irreversible actions, and the existence of real caller consent before analysis begins. A version of this system without either of those is not a smaller version of the solution – it is a different and unacceptable one.

### 2.29.1 The MVP Core (Non-Negotiable)

1. **Real-time voice or text capture on at least one channel**, supporting Hindi, English, and natural Hindi-English code-switching.
2. **Real speech-to-text transcription with word-level timestamps** – no pre-scripted or staged transcript content.
3. **All eight independently-scored risk dimensions** (Section 2.2). These are included in full at the MVP stage, not cut down to a smaller subset, because the underlying rules-layer architecture makes the eighth dimension no more expensive to add than the first – the cost that scales with time is in the polish layered on top of the scores, not the number of dimensions scored.
4. **The single 0-100 Stress Vulnerability Index**, computed from the eight dimensions using the open, documented weighting formula (Section 2.3).
5. **Low / Moderate / High / Critical categorization**, exactly as the problem statement specifies (Section 2.4).
6. **A basic suggested-actions display**, mapped to the risk category, covering all six support types named in the problem statement.
7. **The human-gate senior-approval state machine** for police intervention, witness protection, and non-medical emergency escalation (Section 2.5) – enforced in code, present from the very first working version, because this is the system’s core ethical claim, not an enhancement to it.

8. **A single, clear, plainly worded upfront consent step** before any recording or analysis begins – even before the fuller granular, mid-call-re-asked consent system (Section 2.8) is built out.
9. **A basic, honest, append-only audit log** of what happened during each call – even before it is upgraded to the tamper-evident, hash-chained version (Section 2.21).
10. **A working agent console** showing the live transcript, the eight risk-dimension bars, the colour-coded SVI badge, and the action buttons the agent actually uses to confirm, override, or escalate.

This list, on its own, fully and literally satisfies every bullet point in the Problem Statement’s Expected Solution Requirements (Section 1.3). That is the intended reading of “MVP” here: not a stripped-down demo of an idea, but a complete, ethically sound, literal answer to exactly what was asked – which everything else in this document then builds on top of, rather than the other way around.

### **2.29.2 Fast-Follow (Built Once the MVP Is Stable, Valuable but Not Load-Bearing)**

- The full Silence Fingerprint, with placement classification and crying/breathing-overlap detection (the MVP uses a simpler duration-only silence flag feeding the dissociation score in the meantime).
- The Explainable “why” evidence lines and per-dimension confidence percentages (the MVP shows the eight scores without the underlying evidence trail or confidence number).
- The full granular, multi-purpose, mid-call-re-asked consent system (the MVP uses one clear upfront consent step).
- The tamper-evident, hash-chained upgrade to the audit log (the MVP log is a plain, honest record without the cryptographic chaining).
- The Case Timeline view and the Administrator Analytics Dashboard (the MVP is focused on a single call, not yet a supervisory or district-level tool).

### **2.29.3 The Wow-Factor Layer (Genuinely Optional, Attempted Only If Time Allows, in Priority Order)**

1. **The Silent SOS** (Section 2.20) – highest priority, because it protects the single most vulnerable subgroup this problem statement is about: victims who cannot safely speak freely at all.
2. **Agent Secondary-Trauma Protection** (Section 2.23) – the most genuinely novel addition, and cheap to build once the caller-facing acoustic layer already exists.
3. **Closing the Referral Black Hole** (Section 2.22) – real-time booking and warm hand-off, which requires a mocked external service and is correspondingly more effort.
4. **A dedicated, polished Accessibility Bridge experience** (Section 2.24) – baseline text-channel support already exists for free, since the lexical scoring engine is channel-agnostic by design; this item is specifically about turning that baseline into a first-class, tested, equally-polished experience rather than an incidental side effect.

This ordering reflects impact per hour of effort, not novelty for its own sake: features that protect the most vulnerable callers, or that cost little to add given what the MVP already built, come before features

that are valuable but comparatively expensive or comparatively less urgent.

## **2.30 On Novelty: What We Invented Versus What We Integrated**

Before presenting further additions, it is worth being precise about a question worth asking of any system in this space: is any of this actually new? The honest answer, stated plainly rather than left for a judge to discover, is that most of the individual techniques used throughout this document are not new. Multi-dimensional risk scoring is established in clinical psychology. Silence as a distress signal is established in speech-pathology research. Duress codes are decades old. Human-in-the-loop review is standard practice in serious clinical AI. Automating a validated wellbeing instrument is a natural extension of existing occupational-health tools, not an invention of a new one.

What is genuinely new is the specific combination and the specific context: no system identified in Section 2.28's review combines rigorous, multi-axis, explainable triage with a code-enforced human safety gate, designed specifically for a population where the institution meant to protect them can itself be the danger. That is integration novelty, not invention novelty, and the distinction matters – claiming otherwise invites a credibility failure the moment a technically literate judge asks a direct question about it.

The four capabilities described below are different in kind from everything before them in this document: as far as we have been able to determine, they do not exist, even individually, in any comparable published system. They are presented separately, and with the same honesty about their limitations as everything else in this document, because a genuinely new idea deserves more scrutiny of its risks, not less.

## **2.31 Cross-Call Pattern Linkage - Community-Level Early Warning**

Every system reviewed in Section 2.28, and this solution as described up to this point, treats each call in complete isolation. No system asks whether separate, unconnected callers are independently describing the same perpetrator, the same location, or a correlated pattern of intimidation within the same time window. A single agent handling one call has no way to see this; a supervisor reviewing calls one at a time does not either.

We propose a privacy-preserving pattern-linkage layer that compares anonymised case descriptors – never caller identities – across calls within a district: features such as a named individual, a village or locality, or a specific method of intimidation described independently by multiple callers. When a correlation crosses a defined threshold, a supervisor – never the system alone, and never automatically – is shown a flag: for example, that three unrelated callers in the same district within the past week independently described fear of retaliation involving the same named individual. This turns individually-scoped triage into a community-level early-warning

signal, which is a capability absent from every system in Section 2.28's comparison, because none of them are designed to look beyond the single caller in front of them.

This capability carries real risk and must be built with a matching level of caution: the descriptors compared must be limited to features that cannot re-identify a specific caller on their own, the matching threshold must be set conservatively to avoid false correlations that stigmatise an innocent third party, and any resulting action is limited to raising a flag for a human review, never to any automated escalation. This is the most operationally sensitive of the four new additions in this section, and we present it as a capability requiring institutional partnership and formal privacy review before any pilot – not a feature suitable for a 36-hour build.

## **2.32 AI as a Live Trauma-Informed Interview Coach**

Every capability described earlier in this document has the system observing and scoring what the agent already does. A different kind of addition is an AI that actively coaches the agent's pacing during the call itself, grounded in established forensic-interview science – open-ended questioning, tolerance for silence, avoidance of leading phrasing – research that already exists and is well validated for eliciting disclosures without re-traumatising the person disclosing them, but which is not, as far as we have found, applied as a real-time, in-call coaching layer anywhere in this space.

Concretely: the moment the Silence Fingerprint detects a likely dissociative pause, a small, unobtrusive on-screen cue suggests to the agent “let this silence continue – do not fill it.” When the risk profile suggests the caller may be describing something they have not fully disclosed, a cue suggests “ask an open-ended question here, not a yes/no one.” This shifts the AI from an observer of the conversation to an active collaborator in how the conversation itself is conducted, directly in service of not re-traumatising the caller through well-intentioned but poorly-paced questioning.

The necessary guardrail: coaching is always a suggestion the agent can see and dismiss, never an instruction the agent must follow, and it never overrides the agent's own judgement or training. An agent who has reason to deviate from a coaching prompt – because they know something about the specific caller or situation the system does not – must be free to do so without friction, exactly as with every other suggestion this system makes.

## **2.33 Outcome Feedback Loop - Closing the Field's Biggest Blind Spot**

Every system identified in our literature review, including the most rigorously validated academic research, measures its own accuracy against expert labels applied after the fact – never against what actually happened to the person afterward. This is the single largest blind spot we found across the entire field: a system can be validated as matching a mental-health professional's judgement perfectly and

still be systematically steering people toward support that does not actually help them, and nothing in the existing literature would catch this.

We propose closing this gap directly: with separate, explicit consent obtained at the time of the original call, the system checks back with the caller some weeks later – distinct from the 72-hour and 7-day safety check-ins described in Section 2.9, which exist to catch deterioration, not to evaluate the system – specifically to ask, in a low-pressure way, whether the support they received actually helped. That real-world outcome, not merely an expert’s contemporaneous label, is fed into a human-reviewed process for recalibrating the SVI weighting scheme (Section 2.3) over time.

The necessary guardrails: this outcome check is opt-in, entirely separate from the safety check-ins, and never used to evaluate or penalise the agent who handled the original call. No weight is ever recalibrated automatically from this feedback – every proposed adjustment is reviewed by the same qualified mental-health professionals and domain authorities already responsible for the SVI weighting scheme, exactly as described in Section 2.3, so that outcome data informs human judgement rather than silently reshaping the model on its own.

## **2.34 Physical Safety Bridge via the Community Health Worker Network**

Every safety net described in this document up to this point – the agent, the 72-hour and 7-day check-ins, supervisor review – is phone-based. This is a hard ceiling shared by every crisis-line system identified in Section 2.28’s comparison: if a Critical-risk caller becomes unreachable by phone, none of these systems have any way to verify their physical safety.

India has an existing, extensive network of community health workers – ASHA and Anganwadi workers – already embedded in rural and semi-urban communities nationwide, whose ordinary duties already include home visits for unrelated health purposes. We propose that a Critical-risk case where phone follow-up fails or is judged unsafe can, with the caller’s prior consent and under the same senior-approval gate that governs police and witness-protection actions (Section 2.5), be routed to the already-present local ASHA or Anganwadi worker for a discreet, in-person welfare check – framed to anyone observing it as a routine health visit, never announced as connected to the original call. This closes the phone-only ceiling shared by every comparable system, using existing government infrastructure rather than proposing anything new to build or fund.

This is the addition that requires the most caution of the four, and we say so directly. A poorly handled welfare visit risks alerting a perpetrator that the victim sought help, which could itself cause harm – precisely the failure mode this entire document is built to avoid elsewhere. For this reason, this capability is presented as a documented, high-value direction for institutional partnership and careful protocol design with the relevant state health authorities, not as a feature to be built or demonstrated in a hackathon setting.

## **2.35 Where These Four Additions Fit**

None of the four capabilities in this section belong in the MVP (Section 2.29.1) or the wow-factor layer (Section 2.29.3): each requires either formal privacy review, institutional partnership with a state health department, or a level of real-world testing that a 36-hour build cannot responsibly provide. We present them as a fourth, explicitly separate tier – a frontier layer, beyond both the MVP and the stretch goals, intended to be pursued during the pilot phase described in this document’s broader rollout, not attempted live in front of a judging panel. Presenting genuinely new ideas honestly, including exactly why they are not yet ready to demonstrate, is worth more to this document’s credibility than quietly folding them in as if they were as build-ready as everything that precedes them.